H_2S Scrubbing IsoFlask® with Silver Nitrate – **SAMPLING INSTRUCTIONS**

Included in sampling kit:

- H_2S Scrubbing IsoFlask with Silver Nitrate
- IsoBag® – 1 per IsoFlask
- 60 cc syringe with Luer valve and needle – qty (1)
- Luer septum – qty (2)
- Spare syringe and septa

WARNING: Hydrogen sulfide (H_2S) is a highly toxic gas even at low concentrations. All necessary precautions associated with the handling of samples containing H_2S must be taken prior to using this product.

1. Attach a Luer septum to the IsoFlask®. The septum is press-fit and does not need to be threaded.
2. Use the syringe to draw up 60 cc of source gas.
   - ➢ The source container should be equipped with a septum port and should be under little to no pressure. Pressurized sources should be regulated to less than 5 psig if possible.
   - ➢ Always close the Luer valve before removing the sampling needle from the source container to avoid contaminating the sample and/or releasing H_2S into the atmosphere.
   - ➢ **If the syringe is exposed to excess pressure**, the plunger will be ejected from the end of the syringe forcefully, causing the potential for injury and the release of toxic gas into the atmosphere.
3. Insert the sampling needle into the Luer septum, open the valve, and inject the sample gas into the IsoFlask.
   - ➢ Keep the neck of the IsoFlask up-right during injection. The flask may be rested on its side while not in use, but do not turn it upside down to avoid getting the scrubbing solution into the integrated valve of the IsoFlask.

4. Repeat up to 5 more times (up to 300 cc total injected). **The IsoFlask scrubbing solution can handle up to 300 cc of sour gas with a concentration of up to 30% H_2S.**
   - Note that if the gas source is pressurized, each injection may be in excess of 60 cc when expanded to atmospheric pressure in the IsoFlask. 5 psi of pressure will deliver an additional 20 cc of gas with each injection.
5. Once all gas is injected into the IsoFlask, shake the flask for about 30 seconds to ensure the scrubbing reaction goes to completion.
6. Attach a second Luer septum to the IsoBag.
7. Use the syringe to extract 60 cc of gas from the IsoFlask and transfer to the IsoBag.
   - Continue to keep the neck of the IsoFlask up-right, as it is important that none of the scrubbing solution is introduced into the IsoBag. If the syringe is contaminated with scrubbing solution, re-inject any gas in the syringe back into the IsoFlask and purge the syringe with air several times before continuing.
8. Repeat extraction until only a small amount of gas remains in the IsoFlask.
   - Stop extracting before all gas is removed from the IsoFlask to avoid extracting any of the scrubbing solution. A few mL of gas loss is acceptable.
9. Remove the Luer septa from the IsoFlask and IsoBag.
10. Ship the filled IsoBag and used IsoFlask back to Isotech for analysis and disposal.
    - If the syringe is to be re-used for further sampling, purge the syringe with air before proceeding with the next sample.

60 cc syringe with Luer valve in the CLOSED position.

IsoFlask and Luer septum.

IsoFlask with Luer septum installed properly.

Standard IsoBag.

IsoBag with Luer septum installed properly.